

Experiment 1 — Characterization of Effluent: Physical Parameters

Aim

To characterise a tannery effluent stream by measuring its physical parameters (pH, temperature, conductivity, alkalinity) and its solids content (total, dissolved, suspended and volatile solids), and to compare the streams across the tannery.

Principle / Theory

pH ($= -\log[H^+]$) shows how acidic or alkaline the stream is; tannery streams range from alkaline lime liquor (pH 11–12) to acidic pickle/tan liquor (pH 2.5–4). Conductivity is proportional to dissolved ions and gives a quick TDS estimate ($TDS \approx 0.65 \times EC$). Alkalinity is the acid-neutralising capacity, found by titrating with standard acid to pH 8.3 and pH 4.5.

Solids are the residue left after evaporation. Drying at 105 °C gives the total/dissolved/suspended residue; ignition at 550 °C burns off the organic (volatile) part. TSS is retained on a glass-fibre filter, TDS passes through it, and $TS = TDS + TSS$.

Significance

These are the first numbers any ETP needs. They reveal which stream carries which problem (alkalinity and solids in lime liquor, salt in pickle, etc.) and set the dose for neutralization and the load on the screens and clarifier.

Group Allocation — Five Groups

Each group is given a different tannery stream and runs both parts (physical parameters and solids) on it. Pooling the five streams builds a complete stream-wise characterization of the tannery.

Group	Assigned task	Members
Group 1	Soak liquor (soaking)	_____
Group 2	Lime liquor (liming / unhairing)	_____
Group 3	Spent chrome / tanyard liquor	_____
Group 4	Composite raw effluent (ETP inlet)	_____
Group 5	Treated effluent (ETP outlet)	_____

Apparatus & Instruments

- Multi-parameter meter (pH, temperature, conductivity)
- Burette, pipettes, conical flasks
- Hot-air oven (105 °C); muffle furnace (550 °C); analytical balance; desiccator
- Evaporating dishes, glass-fibre filters with filtration assembly, water bath

Reagents

- Standard 0.02 N H₂SO₄; phenolphthalein; methyl-orange
- Buffer solutions and conductivity standard (calibration)
- Distilled water (solids work needs no reagent)

Procedure

Part A — Physical parameters

1. Calibrate the meter; record pH, temperature and conductivity of your stream; estimate TDS = $0.65 \times \text{EC}$.
2. Alkalinity: to 50 mL sample add phenolphthalein; titrate with 0.02 N H₂SO₄ to colourless (P). Add methyl-orange; titrate to orange-red (T).

Observations

A. Your own group's data

Physical parameters (your stream)

Parameter	Reading 1	Reading 2	Mean
pH			
Temperature (°C)			
Conductivity (µS/cm)			
Estimated TDS (mg/L)			
Titre to pH 8.3, P (mL)			
Titre to pH 4.5, T (mL)			
Sample volume (mL)			

B. Pooled class results (copy every group's result)

Stream	pH	EC ($\mu\text{S}/\text{cm}$)	Total alkalinity	TS (mg/L)	TSS (mg/L)
Soak					
Lime					
Chrome / tanyard					
Composite raw					
Treated outlet					

Calculations

Total alkalinity (mg/L as CaCO_3) = $(T \times N \times 50,000) / V_{\text{sample}}$ (N = 0.02, V = 50 mL)

Precautions

- Calibrate the meter just before use; keep probes wet.
- Use constant-weight drying; cool only in a desiccator.
- Mix the sample thoroughly before each aliquot.